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Professor Vanni Noferini
Editor-in-Chief
The Electronic Journal of Linear Algebra

Dear Professor Noferini,

Please consider my manuscript, “The General Subgroup Permanent-Dominance Conjecture in Order Four,” for publication in *The Electronic Journal of Linear Algebra*.

The manuscript proves the complete order-four case of the general subgroup permanent-dominance conjecture: for every subgroup $H \leq S_4$, every irreducible complex character χ of H , and every 4×4 Hermitian positive-semidefinite matrix A , it establishes $d_\chi^H(A)/\chi(1) \leq \text{per } A$. This goes beyond the usual S_4 immanant specialization by treating all thirty-seven irreducible-character cases arising from the eleven conjugacy classes of subgroups of S_4 .

The main conceptual contribution is the treatment of the two non-real A_4 characters. Their inequalities are reduced to polynomial nonnegativity on the cone of 3×3 positive-semidefinite Gram matrices. A rank-one sum-of-squares identity, an exact interior certificate, and closure prove the required positivity. The remaining thirty-five cases are resolved by general matrix inequalities. A companion Lean formalization checks the central identities and the complete finite reduction.

The general subgroup conjecture was previously known only through matrix order three. The present result closes the next order completely, and the Gram-cone reduction supplies a geometric mechanism that may be useful for related matrix inequalities. The result and its methods lie directly within the matrix-analysis and linear-algebra scope of the journal.

The manuscript is original, has not been published previously, and is not under consideration by another journal. The cited repository contains the preprint and companion formalization. No external funding was received, and I have no competing interests to declare.

Thank you for considering the manuscript.

Sincerely,

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